Multidimensional Evaluation of LLM-Based Coding Agents Beyond Functional Test Metrics

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Research in Progress
Student Challenge

99,7–100,0 %	79,3–92,9 %	3.870–7.489
visible tests	Hidden Tests	Tokens/Test Case

Visible tests show a ceiling effect - the additional metric layer reveals the differences.

1 Study Design & Metrics

288 paired runs
3 agents · 4 tasks × 4 variants
3 prompt levels · 2 modes

Evaluation Pipeline

Configuration · Coding Agent · Code Artifact

Tests · SLOC / MI · AST · Efficiency

Quality Profile

Metric Layer · Six Perspectives

Functionality	visible tests
Robustness	hidden edge cases
Maintainability	SLOC · Complexity · MI
Minimality	appropriate code size
Constraint	rules · test file
Efficiency	runtime · tokens

Qualitative Overall Score

$$Q = \frac{\text{Interface} + \text{Constraint} + \text{Minimality} + \text{Robustness}}{4}$$

* Interface = interface specification met.

Token Normalization
Prompt/Cache + Completion + Reasoning

Statistics
Friedman/Cochran-Q · Wilcoxon/McNemar · Holm

Fair Artifact Basis: n = 70
n = 70: artifacts from all three agents.
n = 96 per agent: full operational evaluation.

2 Key Results

Visible Tests: Ceiling

Claude 100,0 % · Gemini 100,0 % · Codex 99,7 %
global p = 0,368

Hidden-Pass-Rate

higher is better

Claude	92,9 %
Gemini	83,3 %
Codex	79,3 %

global p = 0,0135

Tokens per passed test case

lower is better

Codex	3.870
Claude	6.998
Gemini	7.489

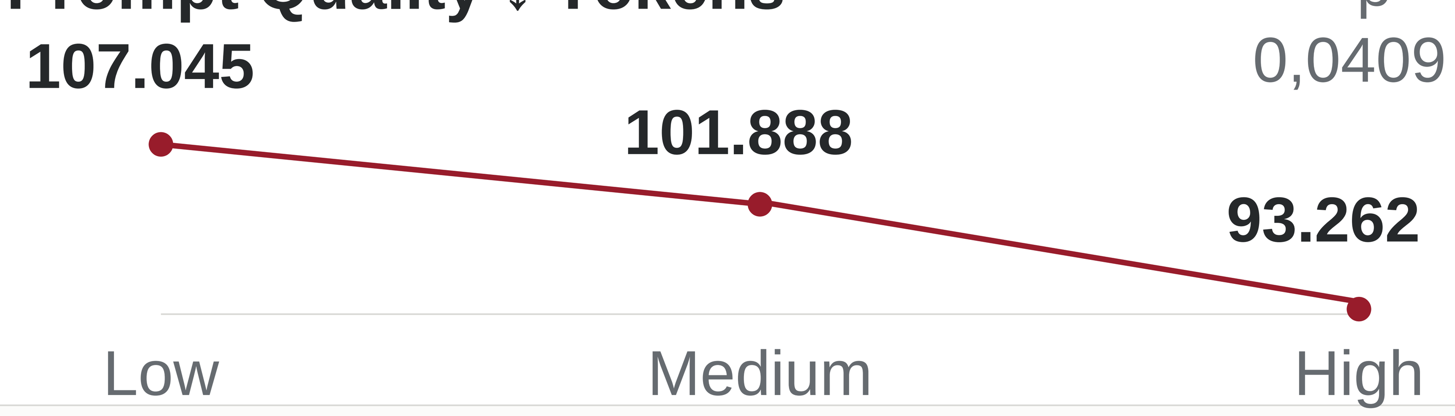
global p < 0,001

Qualitative Overall Score

Claude	0,972
Codex	0,942
Gemini	0,881

all pairwise comparisons significant after Holm correction

Prompt Quality ↓ Tokens p = 0,0409



3 Hypotheses & Conclusion

Hypothesis Testing

H1 Token Saturation — ACCEPTED
More tokens do not explain higher performance. p = 0,913; Hidden-Pass-Rate: p = 0,476

H2 Self-Refining — REJECTED
No robustness gain: Direct 85,6 % vs. Self-Refining 84,8 %; p = 0,179

H3 Prompt Quality — ACCEPTED
Token use decreases: 107.045 → 101.888 → 93.262; p = 0,0409

H4 Prompt Quality — REJECTED
No additional quality gain; p = 0,990

No Universal Ranking

Claude highest quality · Hidden-Pass 92,9 %
Codex best efficiency · lowest complexity
Gemini operationally stable · Compliance 75,7 %

TAKE-AWAY

FUNCTIONAL CORRECTNESS IS NO LONGER ENOUGH.

Evaluation priorities in software engineering are shifting from functional correctness toward quality, robustness, and efficiency.

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